

SAURABH SINGH

ML Engineer & Full-Stack Developer · AI Agents · Automation · FastAPI · React

Ahmedabad, Gujarat

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PROFESSIONAL SUMMARY

ML engineer and full-stack developer with production experience building AI systems that deliver real-world impact — from real-time fault detection pipelines to intelligent crop disease detection apps. Passionate about designing AI agents and automated workflows that connect people, processes, and data. Proficient in Python, FastAPI, React, and n8n, with a track record of shipping end-to-end systems with measurable outcomes.

EXPERIENCE

Data Scientist Intern · Infosys Springboard · Ahmedabad, India

Sep 2025 – Nov 2025

- Built Circuit Guard — an ML system classifying 5+ electrical fault types (short circuit, overload, ground fault) from sensor data with 90%+ accuracy, reducing manual inspection effort by ~60%
- Designed a real-time sensor ingestion and alert pipeline that flags unsafe conditions and recommends corrective actions automatically — mirroring agentic AI decision loops
- Implemented fault classification using Python and scikit-learn (Random Forest, SVM); packaged predictions via FastAPI for downstream integration
- Iterated model performance with a cross-functional team, presenting findings and refining outputs based on stakeholder feedback

PROJECTS

FasalRakshak — AI Plant Disease Detection Agent · Python, TensorFlow/Keras, FastAPI, React [GitHub ↗](#)

- Built a CNN-based model detecting 15+ crop diseases from leaf photos (~90% accuracy) that auto-generates actionable care tips and medicine recommendations — a complete AI agent input-to-output loop
- Developed full-stack system: FastAPI backend serving low-latency predictions as a REST API; React frontend for image upload and results display, optimised for mobile field use by farmers
- Separated model serving, business logic, and UI layers for scalability — applying production-grade AI agent deployment architecture

Circuit Guard — Real-Time Electrical Fault Detection · Python, scikit-learn, FastAPI [GitHub ↗](#)

- Engineered an AI decision agent processing live sensor readings, classifying fault type, and recommending corrective actions — a fully automated safety monitoring pipeline
- Trained and evaluated multiple ML models with cross-validation; achieved 90%+ accuracy across 5+ fault categories

Automated Data & Job Workflow — Enterprise Automation Pipeline · n8n, Google Drive API, Google Sheets API [GitHub ↗](#)

- Designed multi-step agentic workflows in n8n integrating Google Drive and Google Sheets for automated data extraction, transformation, and delivery — directly aligned with Corp8.ai's unified AI workspace model
- Created exportable JSON workflow templates enabling repeatable, trigger-based pipelines (file uploads, scheduled events) with zero manual intervention

Movie Recommender System — AI Recommendation Engine · Python, Streamlit, scikit-learn, TMDb API [GitHub ↗](#)

- Built a content-based recommendation engine using cosine similarity; deployed as a Streamlit web app with live TMDb API integration for posters, genres, and trailers
- End-to-end ML pipeline: data preprocessing, feature engineering, similarity computation, and API integration — shipped as a browser-accessible production app

TECHNICAL SKILLS

AI / ML	Python, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Model Evaluation, Feature Engineering
AI Agents	n8n, Workflow Automation, API Integration, Real-time Pipelines, Agentic System Design
Backend	FastAPI, REST APIs, Python, C++
Frontend	React, JavaScript, HTML, CSS, Streamlit
Tools	Git, GitHub, Google Drive API, Google Sheets API, TMDb API

EDUCATION

B.Tech — Computer Science (AI & ML) · **Adani University** · Ahmedabad, Gujarat

Expected May 2027

- Relevant coursework: Machine Learning, Deep Learning, Data Science, Web Development, Data Structures & Algorithms
- Active member, Machine Learning Branch — Coding Club; led peer learning sessions on ML model building and Python best practices